

Lifecycle category	Practice (consolidated)	Contributing papers	Unified description	Impact factor(s) - RQ1	Measurement / metric - RQ2	Technique tag (optional)
Data	Data-centric reduction & curation	P21, S3	Reduce dataset size / redundant samples and improve data quality/curation to cut training energy with little accuracy loss.	Energy, carbon	Measured training energy vs dataset size	data pruning; curation
Model design	Model pruning & compression	S5, P25, P15, P26, P23, S15	Remove redundant weights/parameters (pruning) or otherwise compress the model to reduce compute and energy.	Energy	FLOPs; measured energy	pruning; compression
Model design	Efficient / sparse architectures	S20, S21, S4, P23, P26	Adopt sparse or Mixture-of-Experts and inherently efficient architectures instead of dense over-parameterised ones.	Energy, carbon	kWh, CO2e; FLOPs	sparsity; MoE
Model design	Efficient model / algorithm selection	P20, S13, S14, S6	Select the smallest / most energy-efficient model or algorithm meeting the task's accuracy needs, informed by comparative energy metrics.	Energy	Accuracy-vs-energy; FLOPs; energy index (S14)	model selection
Training	Efficient hyperparameter configuration & search	S1, P7, S2, P15	Tune training hyperparameters (learning rate, batch size) and use efficient HP/NAS search; poor settings can cost up to ~5x the energy for the same accuracy (S1).	Energy, carbon	Measured energy; GPU-hours	HP tuning; efficient NAS
Training	Predict-and-stop unpromising runs	S10, S9	Forecast a run's energy/carbon before or early in training and stop unpromising runs to avoid wasted compute.	Energy, carbon	Predicted kWh / CO2e (Carbontracker)	early stopping
Training	Efficient fine-tuning	S24, P15	Account for and reduce fine-tuning energy (frequent, done by many actors), not only pre-training cost.	Energy, carbon	Measured fine-tuning energy/carbon	fine-tuning
Training	Compare & select energy-efficient training settings	S6, P26	Empirically compare optimisers/optimisations and pick the most energy-efficient configuration.	Energy	Measured energy of optimisations	—
Inference/serving	Efficient runtime / framework / format choice	P10, P11, S12	Choose the runtime infrastructure, DL framework and model format that minimise inference energy for the target deployment.	Energy	Measured inference energy; battery usage (S12)	runtime/framework choice
Inference/serving	Right-size models for deployment	P5, S11, S13, S25	Match model size/complexity to the deployment context (cloud/edge/mobile/MCU); avoid over-provisioned models at inference.	Energy, carbon	Per-inference energy; accuracy-vs-resource	right-sizing; TinyML
Operational/infrastructure	Carbon-aware scheduling (region & time)	S18, S22, S20, S8, S9, S10, P23, P28	Shift/delay temporally-flexible workloads to lower-carbon times and select low-carbon regions / cloud instances.	Operational carbon	Regional/marginal carbon intensity; CO2e	carbon-aware scheduling
Operational/infrastructure	Efficient datacentre, hardware & accelerators	S20, S21, S15, S7, S4, P23, P28	Use efficient datacentres (low PUE) and energy-efficient hardware/accelerators (GPU/TPU/FPGA) suited to the workload.	Energy, operational carbon	PUE; measured energy; hardware comparison	efficient hardware
Operational/infrastructure	Renewable / low-carbon energy sourcing	P23, S8, S9	Power training/inference with renewable or low-carbon electricity where possible.	Operational carbon	Grid carbon intensity	renewables
Operational/infrastructure	Account for embodied carbon & hardware lifecycle	S4, S23, S25	Include hardware-manufacturing (embodied) carbon and e-waste in the footprint, not only operational energy.	Embodied carbon; e-waste	Lifecycle CO2e accounting (LCA)	embodied; LCA
Cross-cutting	Measurement, reporting & transparency	P9, S8, S10, S9, S17, S14, S23, S2	Treat efficiency/footprint as a first-class, reported metric; use trackers/calculators and standardised reporting.	Energy, carbon	experiment-impact-tracker, Carbontracker, ML CO2 calc, energy index, FLOPs	measurement/reporting
Cross-cutting	Governance & sustainable-by-design	S16, S19, P29, P30, P23	Embed sustainability into AI governance and design principles; use policy levers and research agendas.	(all)	(qualitative)	governance
Cross-cutting	Question the necessity of scale	S3, P9, S23	Challenge the default of ever-larger models/data; prefer sufficiency ('Green vs Red AI').	Compute / efficiency	FLOPs; compute trends	sufficiency

Impact factor	Definition	Contributing papers
Energy (electricity, kWh)	Electricity consumed by compute during training and/or inference.	P5, P7, P10, P11, P15, P20, P25, P26, S1, S5, S6, S7, S11, S12, S13, S14, S17, S24
Operational carbon (CO2e)	Greenhouse-gas emissions from the electricity used to run computation.	S2, S8, S9, S10, S18, S19, S20, S21, S22, S23, S24, P23
Embodied carbon	Emissions from hardware manufacturing / lifecycle, independent of use-phase electricity.	S4, S23, S25
E-waste / resource use	Electronic waste and material/resource use over the hardware lifecycle.	S25, S4
Water footprint	Water used for datacentre cooling and electricity generation.	S4;
Compute / efficiency (proxy)	Compute (e.g. FLOPs/FPO) used as a hardware-agnostic proxy for cost/energy.	P9, S13, S20
Financial cost (proxy)	Monetary cost of hardware/compute, correlated with energy.	S2, P9
Equity / inclusivity (secondary)	High compute cost as a barrier to research participation ('Red vs Green AI').	S2, P9

Metric / Tool	What it captures	Contributing papers	Notes
Energy (kWh, measured)	Direct measured energy use of training/inference.	P7, P10, P11, S1, S5, S6, S7, S12, S13, S14	Most reliable; needs instrumentation
Carbon (CO2e)	GHG emissions from energy use.	S2, S8, S9, S10, S20, S21, S22, S23, S24	Depends on grid carbon factors
FLOPs / FPO (proxy)	Hardware-agnostic compute as an efficiency proxy.	P9, S13	Often uncorrelated with real energy (caveat)
Carbon intensity (regional / marginal)	gCO2 per kWh of the grid, by region and time.	S22, S8, S9, S10, S18	Marginal (S22 Dodge) vs average
PUE	Datacentre energy-overhead multiplier.	S2, S20	Multiplier on compute energy
experiment-impact-tracker (tool)	Energy + carbon reporting framework/tool.	S8	Open-source
Carbontracker (tool)	Predict + track training energy/carbon.	S10	Proactive (stop early)
ML CO2 Impact Calculator (tool)	Estimate carbon from hardware/region/time.	S9	Estimation
Energy consumption index	Standardised index to compare model energy.	S14	Comparative across architectures/GPUs
Energy-estimation models	Estimate ML energy without direct metering; locate hotspots.	S17, S11	Estimation; hotspot identification
Battery usage (on-device)	Battery drain of on-device inference.	S12	Mobile/edge proxy for energy

Finding / tension	Description	Papers
Accuracy-energy trade-off is central	Most practices navigate a trade-off between accuracy/performance and energy/carbon; Pareto-style optimisation recurs.	P5, S20, S13, S1
FLOPs/FPO ≠ real energy	Compute proxies are convenient but often poorly correlated with measured energy; direct measurement is preferred.	P9, S13, S17
Operational vs embodied carbon	Reported footprints are dominated by operational energy, but embodied (manufacturing) carbon and e-waste are significant and often omitted.	S4, S23, S25
Carbon-aware computing is broadly supported	Region/time shifting and low-carbon sourcing are widely endorsed operational levers.	S18, S22, S20, S8, S10
Data-centric vs model-centric levers	Energy can be cut by reducing/curating data as well as by changing the model - complementary levers.	P21, S3, S5, P25
Technical vs governance practices	The corpus spans concrete technical practices and higher-level governance / sustainable-by-design and policy.	S16, S19, P29, P30
Measurement is the enabler and unreliable	Reporting/tooling underpins every other practice, yet estimates carry substantial uncertainty (grid factors, instrumentation, proxies).	S8, S10, S9, S17
Question the necessity of scale	A recurring meta-practice: challenge default scaling; 'Green vs Red AI'.	S3, P9, S23